

An alternative mechanism for recent volcanism on Mars

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[1] Images of the Martian surface indicate that volcanism is a recent process on Mars with ages of the youngest lava flows of about 2 Ma. Interestingly, these young lava flows are confined to the large, old volcanic centers of Tharsis and Elysium, which exhibit a thickened crust compared to surrounding areas. The origin of the volcanic centers but also their longstanding volcanic activity is often explained by strong mantle plumes arising from the core-mantle boundary. However, their existence and sustainment during the last few billion years is uncertain under Martian conditions. Alternatively, the thermal insulation by the crust could cause the longstanding volcanic activity. We show that a locally thickened crust, which has a lower thermal conductivity and is enriched in radioactive heat sources in comparison to the mantle, leads to significant lateral temperature variations in the upper mantle sufficient to generate partial melt even in the present Martian mantle. **Citation:** Schumacher, S., and D. Breuer (2007), An alternative mechanism for recent volcanism on Mars, *Geophys. Res. Lett.*, 34, L14202, doi:10.1029/2007GL030083.

1. Introduction

[2] High resolution images of Martian volcanoes, e.g., by Mars Express have shown that some of them exhibit nearly uncratered and, therefore, extremely young lava flows. Analysis by Neukum *et al.* [2004] leads to the conclusion that some of these flows are as young as 2 Ma. Although the evidence for recent volcanism is overwhelming, the underlying mechanism which causes this ongoing volcanic activity is still not understood. Recent thermal evolution models indicate that volcanism may have stopped at least some hundred millions years ago when the Martian interior cooled significantly below the mantle solidus [Hauck and Phillips, 2002; Breuer and Spohn, 2003, 2006]. It has been debated if the volcanism of large volcanic centers like Tharsis and Elysium could have been sustained by a strong mantle plume [e.g., Kieffer, 2003], bringing enough heat from the core-mantle boundary to the upper mantle to cause melting and volcanism within the stagnant lid. Although this hypothesis may be true for early Mars, a mantle plume – if existent at all – is difficult to sustain under recent conditions and would yield just a slight temperature increase of a few tens of K at the base of the stagnant lid (see discussion in Schumacher and Breuer [2006a]). We therefore concentrate on an alternative explanation which does

not involve temperature heterogeneities caused by mantle plumes.

[3] Most thermal evolution models assume a constant thermal conductivity in mantle and crust of $4 \text{ Wm}^{-1}\text{K}^{-1}$. In a recent study, however, we have shown that different thermal conductivities in crust and mantle due to compositional, temperature and pressure variations should not be neglected [Schumacher and Breuer, 2006a, 2006b]. The value of the crustal conductivity for basaltic material is about $2 \text{ Wm}^{-1}\text{K}^{-1}$ [e.g., Clifford and Fanale, 1985], and the temperature- and pressure-dependent thermal conductivity $k(T, P)$ in the mantle [Hofmeister, 1999] is also lower than the earlier used constant value of $4 \text{ Wm}^{-1}\text{K}^{-1}$. As a consequence, the planetary interior – in particular the upper mantle – cools slower: a global partial melt zone at the base of the stagnant lid exists during a substantial period of time and the present temperature of the average mantle is only a few tens of K below the mantle solidus [Schumacher and Breuer, 2006a]. Thus, only small lateral temperature variations are needed to generate partial melt. In contrast, for models with a crustal thermal conductivity larger than $2 \text{ Wm}^{-1}\text{K}^{-1}$ a global partial melt zone disappears already after about one billion years and the present average mantle temperature is more than 100 K below the mantle solidus [see Schumacher and Breuer, 2006b]. Partial melting in the present mantle but also during the last few billion years would therefore require large lateral temperature variations.

[4] In the present study we investigate the effects of lateral variations in crustal thickness and of the thermal conductivity in crust and mantle on the temperature distribution of the stagnant lid. In particular we try to understand how much the crustal root of about 20 km underneath the large Tharsis rise [Montési and Zuber, 2003] influences the upper mantle temperatures and if it can cause the existence of a partial melt zone.

2. Models

[5] We solve a two dimensional heat conduction equation with a 2-D finite elements code (COMSOL Multiphysics) for the temperature distribution in crust and mantle for present-day conditions. Boundary conditions are a constant surface temperature, a constant mantle heat flow (whose value is given by 1-D parameterized mantle convection models) as well as thermal insulation for the left and right boundary. The width of the models is chosen in a way that the temperature distribution close to the sideways boundaries is not influenced by the thickened crust, i.e., temperatures there are the same for models with and without increased crustal thickness.

[6] Models with constant crustal thickness consist of a 4000 km wide and 50 km thick crust which is underlain by a 250 km thick conducting upper mantle. We limit the depth of the model to 300 km, which is the thickness of the

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